



NTH-Premium Test Series 08 Answer Key

1)	1	2	3	4	61)	1	2	3	4	121)	1	2	3	4
2)	1	2	3	4	62)	1	2	3	4	122)	1	2	3	4
3)	1	2	3	4	63)	1	2	3	4	123)	1	2	3	4
4)	1	2	3	4	64)	1	2	3	4	124)	1	2	3	4
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6)	1	2	3	4	66)	1	2	3	4	126)	1	2	3	4
7)	1	2	3	4	67)	1	2	3	4	127)	1	2	3	4
8)	1	2	3	4	68)	1	2	3	4	128)	1	2	3	4
9)	1	2	3	4	69)	1	2	3	4	129)	1	2	3	4
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50)	1	2	3	4	110)	1	2	3	4	170)	1	2	3	4
51)	1	2	3	4	111)	1	2	3	4	171)	1	2	3	4
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54)	1	2	3	4	114)	1	2	3	4	174)	1	2	3	4
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56)	1	2	3	4	116)	1	2	3	4	176)	1	2	3	4
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59)	1	2	3	4	119)	1	2	3	4	179)	1	2	3	4
60)	1	2	3	4	120)	1	2	3	4	180)	1	2	3	4

Q1) Explanation:

Hint: A seconds pendulum is a pendulum whose period is precisely two seconds.

Explanation: A seconds pendulum has a time period of 2 seconds, but it takes only half the time period, or 1 second, to move from one extreme to the other.

Therefore, **Statement I** is false but **Statement II** is true.

Hence, option **(2)** is the correct answer.

Q2) Explanation:

Hint: Recall the progressive wave motion.

Explanation: In a progressive wave motion, particle velocity is not constant at all times. The particle velocity varies as the particles oscillate around their mean positions. The particle velocity is related to the displacement of the particle in the medium, which changes over time as the wave propagates.

During the propagation of sound, the density of the medium is not uniform throughout. In a sound wave, there are regions of compression (higher density) and rarefaction (lower density), which cause the density of the medium to vary as the wave travels. Therefore, both **(A)** and **(R)** are false.

Hence, option **(4)** is the correct answer.

Q3) Explanation:

Hint: Use $T = 2\pi\sqrt{\frac{l}{g}}$.

Step 1: Find the time period of the pendulum.

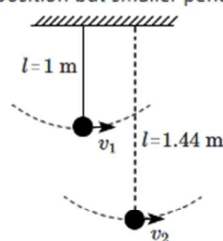
We know that the time period of the pendulum is given as: $T = 2\pi\sqrt{\frac{l}{g}}$

So $T \propto \sqrt{l} \Rightarrow$ for $l = 1 \text{ m}$, $T = T_0$

For $l = 1.44 \text{ m}$, $T = 1.2 T_0$.

Step 2: Compare the oscillation of both the pendulum.

At $t = 0$ sec both pendulums were at the mean position and both in phase. At $t = 1.2 T_0$, bigger pendulum will be at a mean position but smaller pendulum will be toward the right side and it is not in the phase.



Step 3: Find a suitable time.

The suitable time should be multiple of both time periods.

Let $t = 6T_0$.

At this time period, the oscillation of the bigger pendulum will be $= \frac{6T_0}{1.2T_0} = 5$

The oscillation of the smaller pendulum will be $= \frac{6T_0}{T_0} = 6$

So, the oscillations of the bigger pendulum are 5.

Option (1) is correct.

Q4) Explanation:

Hint: $v = \frac{\omega}{k}$

Step 1: Find the required equation of the wave pulse.

The travelling wave pulse is given as;

$$y = \frac{4}{3x^2 + 48t^2 + 24xt + 2}$$

$$\Rightarrow y = \frac{4}{3(x^2 + 16t^2 + 8xt) + 2}$$

$$\Rightarrow y = \frac{4}{3(x+4t)^2 + 2} \quad \dots (1) \quad [(a+b)^2 = a^2 + b^2 + 2ab \cos \theta]$$

Step 2: Find the velocity of the wave.

The general equation of a travelling wave pulse is given by;

$$\Rightarrow y = \frac{a}{(kx + \omega t)^2 + b} \quad \dots (2)$$

Comparing equation (1) and (2) we get;

$$k = 1 \text{ m}^{-1}, \omega = 4 \text{ s}^{-1}$$

The velocity of the wave is given by;

$$\Rightarrow v = \frac{\omega}{k} = \frac{4}{1} = 4 \text{ m/s.}$$

Hence, option (1) is the correct answer.

Q5) Explanation:

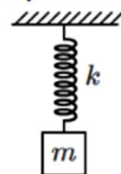
Hint: In a stationary wave, strain is maximum at nodes.

Explanation: At the nodes, the amplitude of the wave is zero, and the strain, which is the ratio of the change in length to the original length of the string, is maximum. At the antinodes, the amplitude of the wave is maximum, and the strain is minimum. Hence, option (2) is the correct answer.

Q6) Explanation:

Hint: $T = 2\pi\sqrt{\frac{m}{k}}$

Explanation: The spring-mass system is shown in the figure.



The time period of the spring-mass system is given by: $T = 2\pi\sqrt{\frac{m}{k}}$.

However, for a spring-mass system, the time period T depends only on the mass of the object and the spring constant. Thus, changes in g (which differ between the equator and poles) do not affect the time period of the spring-mass system. So, if the clock is taken to the pole it will give the correct time because it does not depend on g . Since the time period of the oscillating spring-mass system is independent of gravitational acceleration, the clock will continue to measure time correctly at the poles, just like it did at the equator.

Hence, option (4) is the correct answer.

Q7) Explanation:

Hint: $T = 2\pi\sqrt{\frac{l}{g_{eff}}}$

Step 1: Find the effective acceleration due to gravity inside the liquid.

$$g' = g \left(1 - \frac{\rho}{\sigma}\right)$$

$$g' = g \left(1 - \frac{1}{4}\right) = \frac{3g}{4}$$

Step 2: Find the new time period.

$$\frac{T'}{T} = \sqrt{\frac{g}{g'}}$$

$$= \frac{2}{\sqrt{3}}$$

$$T' = 2 \text{ s}$$

Hence, option (3) is the correct answer.

Q8) Explanation:

Hint: The frequency depends only on the source.

Explanation: When a sound wave is transmitted from air into water, the wave speed will increase as the wave crosses the boundary into the water causing the spacing between crests (the wavelength) to increase because crests move away from the boundary faster than they move up to the boundary. However, the frequency (number of crests passing each second) will be unchanged, since a crest moves away from the boundary every time a crest arrives at the boundary.

Hence, option (4) is the correct answer.

Q9) **Hint:** The x-coordinate, as well as the y-coordinate, are periodic in nature.

Step 1: Find the relation between the x and y.

Given,

$$x = A \cos^2 \omega t$$

$$y = A \sin^2 \omega t$$

On adding both equations,

$$x + y = A$$

It is a straight line between x and y.

Step 2: Find the type of motion of the particle.

As both x and y are periodic in nature, the particle is making an oscillatory motion.

$$\text{Also, } x = A \cos^2 \omega t$$

$$x = \frac{A}{2} + \frac{A}{2} \cos 2\omega t$$

It is an equation of SHM. So the particle is doing SHM along $x + y = A$.

Q10) Explanation:

Hint: Compare the equation with the standard SHM equation.

Step 1: Find the angular frequency of the given SHM.

$$16 \frac{d^2y}{dt^2} + 9y = 0$$

$$\Rightarrow \frac{d^2y}{dt^2} = -\left(\frac{3}{4}\right)^2 y \dots (1)$$

We know that the acceleration of the SHM is related to angular frequency by the relation below;

$$a = -\omega^2 y \dots (2)$$

From equations (1) and (2) we get;

$$\Rightarrow \omega = \frac{3}{4} \text{ s}^{-1}$$

Step 2: Find the time taken by the particle to reach half of its amplitude.

The displacement of the particle on reaching half the amplitude is $A/2$, where A is the amplitude of the given SHM. Therefore, the time taken to reach half the amplitude is given by;

$$A/2 = A \sin(\omega t)$$

$$\Rightarrow \frac{1}{2} = \sin(\omega t)$$

$$\Rightarrow \omega t = \frac{\pi}{6} \quad [\text{multiply } \sin^{-1} \text{ both sides}]$$

$$\Rightarrow t = \frac{\pi}{6\omega}$$

$$\Rightarrow t = \frac{2\pi}{9} \text{ s}$$

Hence, option **(4)** is the correct answer.

Q11) Explanation:

$$\text{Hint: } \lambda = \frac{2\pi}{k}$$

Step: Analyse each statement one by one.

The equation of the wave is given by;

$$y = A \sin 314 \left[\frac{t}{0.5 \text{ s}} - \frac{x}{100 \text{ m}} \right]$$

$$\Rightarrow y = A \sin \left[\frac{314t}{0.5 \text{ s}} - \frac{314x}{100 \text{ m}} \right]$$

$$\Rightarrow y = A \sin[628t - 3.14x] \dots (1)$$

The general equation of the wave is given by;

$$y = A \sin(\omega t - kx) \dots (2)$$

From the equation (1) and (2) we get;

$$\omega = 628, k = 3.14$$

$$\Rightarrow 2\pi n = \omega = 628$$

$$\Rightarrow n = \frac{628}{2\pi} = 100 \text{ Hz}$$

The wavelength of the wave is given by;

$$\lambda = \frac{2\pi}{k}$$

$$\Rightarrow \lambda = \frac{2 \times 3.14}{3.14}$$

$$\Rightarrow \lambda = 2 \text{ m}$$

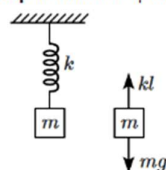
Therefore, the correct statements are **(B)** and **(C)** only.

Hence, option **(3)** is the correct answer.

Q12) Explanation:

Hint: $T = 2\pi\sqrt{\frac{m}{k}}$

Step 1: Find the spring constant.



$$F = kl = mg$$

$$\Rightarrow k = \frac{mg}{l}$$

Step 2: Find the time period of oscillation.

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$= 2\pi\sqrt{\frac{l}{g}}$$

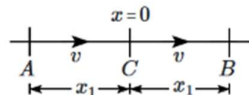
Hence, option (1) is the correct answer.

Q13) Explanation:

Hint: Identify the total time for SHM.

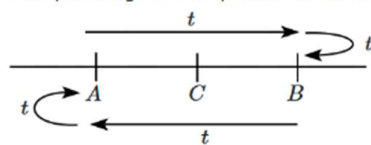
Step 1: Identify the location of points A and B.

Let us take points A and B equidistance from the mean position C. The points A and B are moving with the same velocity $\vec{v}$ given by expression $\vec{v} = \omega\sqrt{A^2 - x^2}$.



Step 2: Find the time taken by the particle for the motion $A \rightarrow B$ and $B \rightarrow A$.

As the particle is moving from point A to B, the time taken by the particle is t . Then the particle goes from point B to extreme and comes back to B, time taken by the particle is t . Due to the symmetry of SHM time taken by the particle to go from B to A is t . Then the particle goes from point A to extreme and comes back to A, the time taken by the particle is t .



Step 3: Find the time period of the SHM.

The period of SHM = $t + t + t + t = 4t$.

$\Rightarrow$ The time taken by a particle performing SHM to pass from the point A to B is 2 s, so the time period of SHM = 8 s.

Option (2) is correct.

Q14) Hint: The acceleration is directly proportional to the displacement of the pendulum from the mean position.

Step 1: Find the angular velocity of the pendulum.

$$|\alpha| = \omega^2 x$$

$$\Rightarrow 20 = \omega^2 \times 5$$

$$\Rightarrow \omega^2 = 4$$

$$\Rightarrow \omega = 2 \text{ rad/s}$$

Step 2: Find the time period of the pendulum.

Now,

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi \text{ s}$$

Q15) Explanation:

Hint: $T = 2\pi\sqrt{\frac{l}{g}}$

Explanation: The time period of oscillations is given by: $T = 2\pi\sqrt{\frac{l}{g}}$. As we can see the period depends only upon the length of the pendulum and acceleration due to gravity. It is independent of the amplitude of oscillation. Therefore the time period remains the same upon changing the amplitude.
Hence, option (1) is the correct answer.

Q16) Explanation:

Hint: Beat frequency = 2 Beats/sec

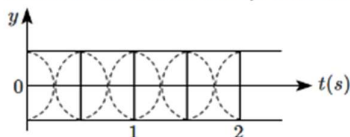
Step: Find the correct wave pattern produced by the superposition of two waves of frequencies 9 Hz and 11 Hz.

The beat frequency is given by;

$$f_b = |f_1 - f_2| = |9 - 11| = 2 \text{ Hz}$$

The beat frequency is 2 Hz means the two waves are interfering in such a way that the resulting combined wave fluctuates in amplitude at a rate of 2 cycles per second.

Therefore, the correct wave pattern showing $f_b = 2 \text{ Hz}$ is shown in the figure below;

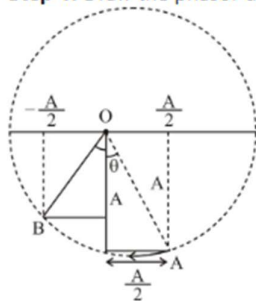


Hence, option (1) is the correct answer.

Q17) Explanation:

Hint: Use a phasor diagram to find the time taken by a particle to reach from $x = \frac{A}{2}$ to $x = -\frac{A}{2}$.

Step 1: Draw the phasor diagram.



Step 2: Find the angle with the help of the diagram.

$$\text{As, } \sin\theta = \frac{A/2}{A} = \frac{1}{2}$$

$$\Rightarrow \theta = 30^\circ$$

$$\Rightarrow \Delta\theta = \frac{\pi}{3}$$

Step 3: Find the time taken by a particle to reach from $x = \frac{A}{2}$ to $x = -\frac{A}{2}$.

$$t = \frac{\Delta\theta}{\omega} \text{ where, } \omega = \frac{2\pi}{T}$$

$$\Rightarrow t = \frac{T}{6}$$

Option (4) is correct.

Q18) Explanation:

Hint: $f = \frac{v}{2l}$

Step: Find the frequency of the fifth harmonic.

The frequency of the n -th harmonic is given by;

$$f_n = n \times f_{\text{fundamental}}$$

The fundamental frequency ($f_{\text{fundamental}}$) of the string is 220 cps.

$$\Rightarrow f_5 = 5 \times 220 = 1100 \text{ cps}$$

Hence, option (3) is the correct answer.

Q19) Explanation:

Hint: Fundamental frequency, $\nu = \frac{v}{2l}$.

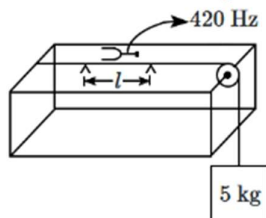
Step: Find the value of the load.

The fundamental frequency of the sonometer wire is given by the relation $\nu = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$

where $\mu = \frac{m}{l}$ is the linear mass density of the wire.

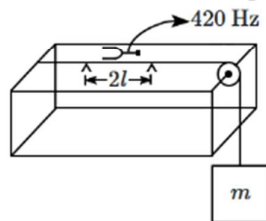
Case 1:

In the first case, the length is l and mass is 5 kg.



Case 2:

In the second case, the length is $2l$ and let mass be m .



If the frequency remains the same in both cases then, $\frac{\nu}{\nu} = \frac{\frac{1}{2l} \sqrt{\frac{T}{\mu}}}{\frac{1}{2l} \sqrt{\frac{T'}{\mu}}} \Rightarrow T' = 4T$

As acceleration due to gravity g remains the same then mass is increased four times.

$$\Rightarrow m = 4 \times 5 \text{ kg} = 20 \text{ kg.}$$

Option (4) is correct.

Q20) Explanation:

Hint: The bob is experiencing a restoring force toward equilibrium.

Step: Analyse the given options.

1. **Kinetic energy:** Kinetic energy keeps changing throughout the oscillation cycle. It is maximum at the mean position when the bob has the highest speed and minimum at the extreme positions where the bob momentarily stops. So, kinetic energy is not conserved.
2. **Momentum:** Momentum is also not conserved as the direction of the bob's velocity keeps changing during the oscillation. At the extreme positions, the momentum is momentarily zero, but it gains a non-zero value in the opposite direction as it starts swinging back.
3. **Potential energy:** Potential energy does change throughout the oscillation cycle. It is maximum at the extreme positions where the bob is farthest from the equilibrium point and minimum at the mean position where the bob is closest.
4. **Total energy:** The change in potential energy is compensated by the opposite change in kinetic energy, making the total energy conserved. Therefore, the total mechanical energy, which is the sum of kinetic and potential energy, remains constant throughout the small oscillations of a simple pendulum.

Hence, option (4) is the correct answer.

Q21) correct option is c

Q22) Explanation:

Hint: $f_b = |f_2 - f_1|$

Explanation: When two nearly equal frequencies are sounded together, the beats are heard. The beat frequency is given by:

$$f_b = |f_2 - f_1| = |102 - 100| = 2 \text{ Hz.}$$

Hence, option (1) is the correct answer.

Q23) Explanation:

Hint: $T = \frac{2\pi}{\omega}$

Step: Find the time period of vertical oscillations of the system.

As the system goes through a vertical oscillation. The length of the rod is l , when it is displaced by the distance x , the spring gets compressed by $(x/2)$ so it applied an additional force in the upward direction which is given by;

$$\Rightarrow F = k \times \frac{x}{2}$$

It applies an additional torque about the point A that is given by;

$$\tau = I\alpha$$
$$- \left(k \frac{x}{2}\right) \left(\frac{l}{2}\right) = ml^2 \alpha$$

Here, $a = l \times \alpha$

$$\Rightarrow a = -\frac{k}{4m}x$$

Compared with the standard equation; $a = -\omega^2 x$, we get;

$$\omega = \frac{1}{2} \sqrt{\frac{k}{m}}$$

The time period of vertical oscillations of the system is given by;

$$T = \frac{2\pi}{\omega}$$

$$\Rightarrow T = 4\pi \sqrt{\frac{m}{k}}$$

Hence, option (3) is the correct answer.

Q24) **Explanation:**

Hint: Express the given function in a single sine function.

Step: Find the value of k .

$$y = \sqrt{2} \left[\sin(\omega t) \cdot \cos\left(\frac{\pi}{4}\right) + \cos(\omega t) \sin\left(\frac{\pi}{4}\right) \right]$$

$$= \sqrt{2} \sin\left(\omega t + \frac{\pi}{4}\right)$$

Angular frequency = ω

$k = 1$.

Option (2) is correct.

Q25) **Explanation:**

Hint: $v \propto \sqrt{T}$

Step 1: Find the value of the temperature T in $^{\circ}\text{C}$.

The speed of sound at the temperature T is $v_1 = 336 \text{ m/s}$.

By comparing given pressure wave ($P = 0.01 \sin[1000t - 3x] \text{ Nm}^{-2}$) with the general equation $P = P_0 \sin(\omega t - 3x)$ we get;

$\omega = 1000 \text{ rad/s}$, $k = 3 \text{ rad m}^{-1}$

Speed of sound at 0° is given by;

$$v_2 = \frac{\omega}{k} = \frac{1000}{3}$$

The velocity of sound is directly proportional to the square root of temperature i.e., $v \propto \sqrt{T}$

$$\frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}}$$

$$\Rightarrow \frac{336 \times 3}{1000} = \sqrt{\frac{T}{273}}$$

$$\Rightarrow \sqrt{\frac{T}{273}} = 1.008$$

Squaring both sides we get;

$$\frac{T}{273} = 1.008 \times 1.008 = 1.016064$$

$$\Rightarrow T = 1.016 \times 273 = 277.385 \text{ K} \approx 277 \text{ K}$$

The temperature in degrees Celsius is given by;

$$T(^{\circ}\text{C}) = T(\text{K}) - 273 \text{ K} = 277 - 273 = 4^{\circ}\text{C}$$

Hence, option (1) is the correct answer.

Q26) **Explanation:**

Hint: $f = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$

Step: Find the velocity of the transverse wave in the string.

The fundamental mode of frequency of the string is given by;

$$f = \frac{1}{2l} \sqrt{\frac{T}{\mu}} = \frac{v}{2l} \dots (1)$$

The length of the string is given by; $\frac{m}{l} = \frac{18}{20} = 0.9 \text{ m}$

The velocity of the transverse wave in the string from the (1) equation is given by; $v = 2lf = 2 \times 0.9 \times 50 = 90 \text{ m/s}$

Therefore, the velocity of the transverse wave in the string is **90 m/s**.

Hence, option (3) is the correct answer.

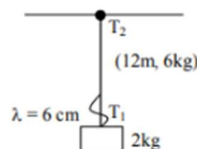
Q27) correct option is b

Q28) Explanation:

Hint: Wavelength is directly proportional to the velocity of the wave.

Step: Find the wavelength of the wavetrain at the top of the rope.

The tension in the rope at different instants is shown in the figure below;



We know that the wavelength of the wave is directly proportional to the velocity of the wave i.e., $\lambda = \frac{v}{f} = \frac{\sqrt{\frac{T}{\mu}}}{f}$ $\left[v = \sqrt{\frac{T}{\mu}} \right]$

$$\Rightarrow \lambda = \sqrt{\frac{T}{\mu}}$$

$$\Rightarrow \frac{\lambda_2}{\lambda_1} = \sqrt{\frac{T_2}{T_1}}$$

$$\Rightarrow \lambda_2 = \lambda_1 \sqrt{\frac{T_2}{T_1}}$$

$$\Rightarrow \lambda_2 = (6\text{ cm}) \times \sqrt{\frac{8g}{2g}} \quad [T_2 = 8g(6g + 2g), T_1 = 2g]$$

$$\Rightarrow \lambda_2 = 6 \times 2 = 12\text{ cm}$$

Hence, option (1) is the correct answer.

Q29) Explanation:

$$\text{Hint: } T = 2\pi\sqrt{\frac{m}{k}}$$

Step: Find the mass of the required block.

Let the mass of the block to be placed on the platform be m' kg.

The time period of a spring mass system is given by;

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$T \propto \sqrt{m}$$

$$\Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{m_1}{m_2}}$$

$$\Rightarrow \frac{3}{6} = \sqrt{\frac{2}{2+m'}}$$

$$\Rightarrow m' = 6\text{ kg}$$

Hence, option (4) is the correct answer.

Q30) Explanation:

$$\text{Hint: } \frac{1}{k_{eff}} = \frac{1}{k_1} + \frac{1}{k_2}$$

Step: Find the effective spring constant.

$$\frac{1}{k_{eff}} = \frac{1}{k_1} + \frac{1}{k_2}$$

$$\frac{1}{k_{eff}} = \frac{1}{k} + \frac{1}{k} = \frac{2}{k}$$

$$\Rightarrow k_{eff} = \frac{k}{2}$$

Hence, option (3) is the correct answer.

Q31) Explanation:

Hint: Recall the concept, the combination of springs.

Step 1: Find the effective spring constant of the system.

Here, two springs are in parallel combination, $k_{net} = 2k$

This combination is in series with the other spring.

The new spring constant is given by; $k'_{net} = \frac{2k \times 2k}{2k + 2k} = k$

Step 2: Find the time period of oscillation.

The time period of the oscillation is given by;

$$T = 2\pi \sqrt{\frac{M}{k'_{net}}}$$

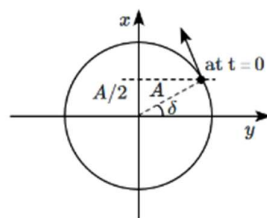
$$\Rightarrow T = 2\pi \sqrt{\frac{M}{k}}$$

Hence, option (1) is the correct answer.

Q32) Explanation:

Hint: Use a phasor diagram.

Step 1: Draw the phasor diagram for the given SHM.



Step 2: Find the phase of the SHM.

In the phasor diagram

$$\sin \delta = \frac{A/2}{A} = \frac{1}{2}$$

$$\delta = \frac{\pi}{6} \text{ radian}$$

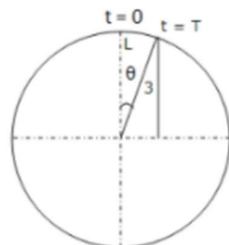
Hence, option (2) is the correct answer.

Q33) **Hint:** Angle made by the radius vector with the y-axis is given by the angular velocity.]

Step 1: Find the angular velocity of the particle.

$$T = 4 \text{ sec}$$

$$\omega = \frac{2\pi}{T} = \frac{\pi}{2} \text{ rad / sec}$$



Step 2: Find the projection on the y-axis.

$$y = A \cos \omega t$$

$$= 3 \cos \left[\frac{\pi t}{2} \right]$$

Q34) $T = 2\pi\sqrt{\frac{l}{g}}$

Step 1: Obtain relation between

T^2 & l

$$T^2 = \left(\frac{4\pi^2}{g}\right) l$$

$y = mx$

So, the graph between T^2 and l will be a straight line that passes through the origin.

Q35) **Explanation:**

Hint: Apply the principle of superposition.

Explanation: To cancel a wave we have to make the amplitude of the wave equal to zero.

To make the amplitude zero, the crest should overlap the trough of the wave and the trough should overlap the crest of the wave.

Hence, option **(1)** is the correct answer.

Q36) **Explanation:**

Hint: $f = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$

Step 1: Find the new linear mass density.

The linear mass density in the string is given by;

$$\mu = \frac{M}{L}$$

$$\Rightarrow \mu = \frac{\rho V}{L}$$

$$\Rightarrow \mu = \frac{\rho AL}{L}$$

$$\Rightarrow \mu = \rho A$$

Hence;

$$\Rightarrow \mu_1 = \rho(\pi r^2)$$

$$\Rightarrow \mu_2 = \rho(4\pi r^2)$$

$$\Rightarrow \mu_2 = 4\mu_1$$

Step 2: Find the frequency.

The frequency of the wave produced on a string is given by;

$$f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

$$\Rightarrow f \propto \frac{1}{\sqrt{\mu}}$$

$$\Rightarrow \frac{f_2}{f_1} = \sqrt{\frac{\mu_1}{\mu_2}}$$

$$\Rightarrow f_2 = \frac{n}{2} \quad [f_1 = n]$$

Hence, option **(3)** is the correct answer.

Q37) Explanation:

Hint: Compare with the equation $y = A \sin(\omega t - kx)$.

Step: Determine the correct match.

(a) On comparing the given equation with the standard equation $y = A \sin(\omega t - kx)$, $\omega = B$, $C = k$. The wave speed is given by;
 $v = \frac{\omega}{k} = \frac{B}{C}$.

(b) Maximum particle speed is given by; $v_{max} = A\omega = AB$.

(c) Wave frequency is given by; $\nu = \frac{1}{T} = \frac{\omega}{2\pi} = \frac{B}{2\pi}$

(d) Wavelength is given by; $\lambda = \frac{2\pi}{k} = \frac{2\pi}{C}$

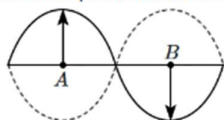
Therefore, the correct match is a - r, b - s, c - p, d - s.

Hence, option (4) is the correct answer.

Q38) Explanation:

Hint: $\Delta\phi = \left(\frac{2\pi}{\lambda}\right) \Delta x$

Step: Find the phase difference.



The phase difference $\Delta\phi$ between two particles at positions A and B is related to their separation by the following equation:

$$\Delta\phi = \frac{2\pi}{\lambda} \times \Delta x$$

If the distance d between A and B is $\lambda/2$, then the phase difference will be: $\Delta\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{2} = \pi$

Thus, for two points separated by half a wavelength, the phase difference is π .

In the absence of specific information about the distance between points A and B , the most typical case is that they are separated by half a wavelength, leading to a phase difference of π .

Hence, option (4) is the correct answer.

Q39) Explanation:

Hint: Use $T = 2\pi\sqrt{\frac{l}{g}}$

Explanation: The time period of SHM is given by $T = 2\pi\sqrt{\frac{l}{g}}$. As we can see the time period is independent of amplitude of oscillation. So the time period remains unchanged.

Option (4) is correct.

Q40) Explanation:

Hint: $v = \sqrt{\frac{T}{\mu}}$

Step: Find the speed at the bottom (v_b) of the rope.

The speed of a transverse wave along a rope is given by;

$$\Rightarrow v = \sqrt{\frac{T}{\mu}}$$

The tension (T_b) is due to the weight of the block (M) hanging from the rope is given by;

$$\Rightarrow T_b = Mg, \text{ where } M = 3m \text{ (the mass of the block is 3 times the mass of the rope).}$$

The tension (T_t) is due to the weight of the block and the entire rope is given by;

$$\Rightarrow T_t = (M + m)g = (3m + m)g = 4mg$$

The speed at any point depends on the tension at that point is given by;

$$\Rightarrow v_t = \sqrt{\frac{T_t}{\mu}}, \quad v_b = \sqrt{\frac{T_b}{\mu}}$$

The ratio of speeds is given by;

$$\Rightarrow \frac{v_t}{v_b} = \sqrt{\frac{T_t}{T_b}}$$

Substitute the known values we get;

$$\Rightarrow T_t = 4mg \text{ and } T_b = 3mg$$

$$\Rightarrow \frac{v_t}{v_b} = \sqrt{\frac{4mg}{3mg}} = \sqrt{\frac{4}{3}}$$

Hence, option (2) is the correct answer.

Q41) Explanation:

Hint: $v = \sqrt{\frac{T}{\mu}}$

Step 1: Find the tension in the rope.

The tension in the rope at a distance x from the bottom is given by; $T = mg = \mu xg$

Step 2: Find the speed of the wave at a distance x from the bottom.

So, the speed of the wave at a distance x from the bottom is given by; $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{\mu xg}{\mu}} = \sqrt{gx}$.

Hence, option (2) is the correct answer.

Q42) **Hint:** To form a stationary wave, waves should travel in the opposite direction.

Step: At $x = 0$, obtain node always.



$$y_1 = a \cos(kx + \omega t)$$

$$y_2 = -a \cos(kx - \omega t)$$

$$\text{At } x=0$$

$$y_1 = a \cos \omega t$$

$$y_2 = -a \cos(-\omega t)$$

$$= -a \cos \omega t$$

$$y_N = y_1 + y_2$$

$$= 0$$

Q43) Explanation:

Hint: $F = -kx$

Step: Find the correct match.

In the presence of air resistance, the bob oscillates with decreasing amplitude. Therefore, the correct option is (a) $\rightarrow$ (IV)

The restoring force of a spring is directly proportional to the displacement of the spring. Therefore the graph is a straight line passing through the origin with a negative slope. Therefore, the correct option is (b) $\rightarrow$ (III)

In case of negligible air resistance, the bob executes SHM with constant amplitude. Therefore, the correct option is (c) $\rightarrow$ (II)

The total mechanical energy of a particle undergoing SHM is constant. Therefore, the correct option is (d) $\rightarrow$ (I)

Therefore, the correct match is (a) $\rightarrow$ (IV), (b) $\rightarrow$ (III), (c) $\rightarrow$ (II), (d) $\rightarrow$ (I).

Hence, option (2) is the correct answer.

Q44) Explanation:

Hint: General form of wave is given by: $y = f(t - \frac{x}{v})$

Step: Identify the equation representing a travelling wave.

$$y = A \sin(15x - 2t)$$

$$\Rightarrow y = A \sin\left(15\left(x - \frac{t}{15}\right)\right)$$

$$\therefore y = f\left(x - \frac{t}{v}\right)$$

Therefore, the equation $y = A \sin(15x - 2t)$, represents a travelling wave.

Hence, option (1) is the correct answer.

Q45) Explanation:

Hint: $T = 2\pi\sqrt{\frac{l}{g}}$

Step: Find the relation between frequency and length of a simple pendulum.

$$T = 2\pi\sqrt{\frac{l}{g}}$$

$$T \propto \sqrt{l}$$

$$f \propto \frac{1}{\sqrt{l}}$$

$$\Rightarrow f^2 l = \text{Constant}$$

Hence, option (3) is the correct answer.

Q46) Explanation:

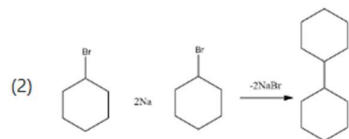
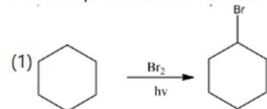
Hint: Second reaction is an example of Wurtz reaction.

Explanation:

(1) In the first reaction, there will be free radical reaction and the Br atom will be attached to the cyclohexane ring.

(2) In second step, Wurtz reaction takes place. It is used to synthesize higher alkanes by a reaction between alkyl halides and metallic sodium in the presence of dry ether.

The complete reaction will be:

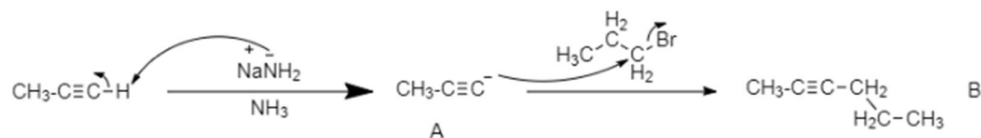


Hence, the correct answer is option 1.

Q47) Hint: Acid-Base reaction followed by substitution reaction

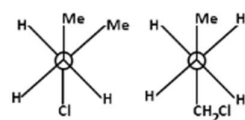
The first reaction is an acid-base reaction. Product A is acetylide. The acetylide acts as a nucleophile and gives a nucleophilic substitution reaction with an alkyl bromide.

The reaction is as follows:



Q48) Explanation:

The given molecules are:



In these structures, the molecular formula is the same but the connectivity of the atoms is different.

(i) Two compounds with the exact same connectivity that are mirror images of each other but that are not identical to each other are called enantiomers. Hence, the given compounds can't be enantiomers as the connectivity of atoms is not the same.

(ii) Positional isomers are constitutional isomers with the same carbon skeleton and will have the same functional group. They differ from each other in the position of the functional groups on the carbon chain and the phenomenon is known as positional isomerism.

The given compounds are position isomers because of the different connectivity of the chlorine atom.

(iii) Conformational isomerism is a type of stereoisomerism in which the isomers can change into each other by rotating around single bonds.

The rotation of single bonds does not interconvert either of these structures. So both are not conformers.

Hence, **option 2** is the correct answer.

Q49) Hint: In the first reaction, alkyne is obtained as product

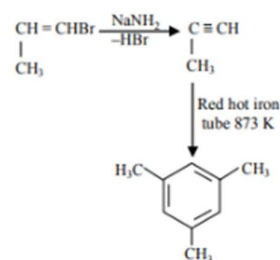
The first reaction is an example of a dehydrohalogenation reaction. This reaction is also known as an elimination reaction.

In this reaction, bromoalkene reacts with sodamide and propyne is obtained as a product.

In the second reaction, propyne derivative on passing through a red hot iron tube at 873K undergoes cyclic polymerization.

Three molecules polymerise to form an alkyl derivative of benzene.

The reactions are as follows:



pseudocumene

Q50) Explanation:

Hint: The numbering will start with the double-bonded carbon atom.

Explanation:

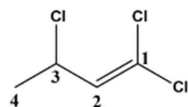
The name of the given compound is: **1, 1, 3 - trichlorobut - 1 - ene**

The suffix "-ene" is used for an alkene that is present at the first carbon. Additionally, the prefix "but-" indicates that there are four carbon atoms in the parent chain.

Three chlorine substituents are present in the compound. Two chlorine atoms are present at the first carbon atom and the third chlorine atom is present at the third carbon atom.

Thus, the correct structure of the given compound is:

1,1,3-trichlorobut-1-ene



Option 2 is the correct choice.

Q51) Hint: As number of carbon atom increases in the group the inductive effect also increases.

As number of carbon increases the inductive effect of alkyl group also increase. The tertiary carbon show more inductive effect than secondary carbon and secondary carbon shows more inductive effect than primary carbon.

Hence, the correct order is $(\text{CH}_3)_3\text{C}- > (\text{CH}_3)_2\text{CH}- > \text{CH}_3\text{CH}_2-$

Q52) Explanation:

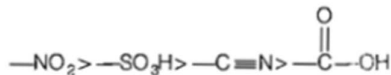
Hint: Rate of reaction depends on the stability of carbocation

Q53) Explanation:

Meta-directing deactivating groups are those that either do not donate electrons by resonance (sulfonic acid groups and ammonium ion groups) or withdraw electrons by resonance (carbonyl, nitrile, and nitro groups).

Meta-directing groups direct electrophilic attack on the "meta" positions of the benzene ring. When an electrophile arrives, it interacts with more electron-density sites and attaches to them, resulting in a meta-product.

The deactivating tendency of given groups follows the order



Thus, $-\text{NO}_2$ is the most deactivating group. Hence, **option 4** is the correct answer.

Q54) Explanation:

Hint: Electron-donating groups significantly increase the ring's electron density.

Explanation:

Sulphonation is an electrophilic aromatic substitution reaction. m-xylene, as it has two methyl groups meta to each other. Since both are activating groups, there could be a greater electron density at the ortho and para positions with respect to other compounds.

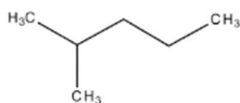
Toluene is also reactive towards sulphonation because the methyl group in toluene donates electrons, making the ring electron-rich.

Chlorobenzene has decreased nucleophilicity due to the chlorine. Similarly, Nitrobenzene has the lowest nucleophilicity due to the nitro group's electron-withdrawing effect.

Hence, **option 4** is the correct answer.

Q55) Explanation:

(i) The structure of 3-methyl pentane is as follows:



(ii) In conformer 1, the front carbon has a CH_3 and two hydrogens. The back carbon has an ethyl group and two hydrogen atoms.

So, it is a simple structure of pentane. It is not 2-methylpentane.

(iii) In conformer 2, the front carbon has a methyl group, a hydrogen atom, and an ethyl group. The back carbon has only a methyl group and two hydrogen atoms.

It is a representation of 3-methylpentane.

(iv) In conformer 3, both front and back carbon have two hydrogen atoms and one methyl group each. This is the correct representation of butane and not 2-methylpentane.

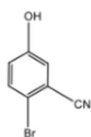
(v) In conformer 4, the front carbon has two methyl groups and a hydrogen atom while the back carbon has an ethyl group and two hydrogen atoms.

This is the correct representation of 3-methylpentane.

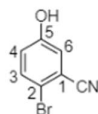
Hence, the correct answer is **option 4**.

Q56) Hint: The CN group gets higher priority among all the groups

The structure of the given compound is as follows:



The CN group gets higher priority than OH and Br hence, the numbering should be done in such a way that CN gets higher priority and the lowest number. The structure with the numbering system is as follows:



Benzene ring containing CN group is known as benzonitrile. The IUPAC name of the compound is 2-Bromo-5-hydroxybenzonitrile

Q57) A compound is said to be aromatic if it satisfies the following three conditions:

- (i) It should have a planar structure and every carbon atom in a ring must be sp^2 hybridised.
- (ii) The π -electrons of the compound are completely delocalized in the ring.
- (iii) The total number of π -electrons present in the ring should be equal to $(4n + 2)$, where $n = 0, 1, 2 \dots$ etc. This is known as Huckel's rule.

Hence, fourth option is the incorrect option.

Q58) Explanation:

$$\text{Hint: } R_f = \frac{\text{Distance travelled by compound}}{\text{Distance travelled by solvent}}$$

Q59) Explanation:

The order of reactivity parallels the order of acidity. The stronger the acid, the more readily it provides H^+ ions for addition to the double bond. The stability of hydrogen halides decreases down the group due to decrease in bond ($H-X$) dissociation energy in the order: $HI > HBr > HCl$

Bond energy of HI is 296.8 kJ/mol, HBr is 362.7 kJ/mol and HCl is 430.5 kJ/mol. As bond energy decreases the reactivity of hydrogen halide compound increases. In the case of hydrogen halide, down the group reactivity increases with propene.

Hence, $HI > HBr > HCl$ is the order of reactivity with propene. So, **option 3** is the correct answer.

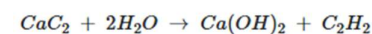
Q60) Hint: Kjeldahl's method sulphuric acid is used as a reagent.

Test with their reagent is as follows:

- | Test | Correct reagent |
|---------------------------------------|--------------------------|
| (i) Lucas test $\rightarrow$ conc. | $HCl + ZnCl_2$ |
| (ii) Dumas method $\rightarrow$ | CuO / CO_2 |
| (iii) Kjeldahl's method $\rightarrow$ | H_2SO_4 |
| (iv) Hinsberg Test $\rightarrow$ | $C_6H_5SO_2Cl + aq. KOH$ |

Q61) Explanation:

(i) Calcium carbide is not volatile or soluble in any known solvent and reacts with water to yield acetylene gas and calcium hydroxide.



(ii) Acetylene, on passing through a red hot iron tube at 873K, undergoes cyclic polymerization. Three of the acetylene molecules polymerize to form benzene.



(iii) Benzene reacts with CH_3Cl in the presence of anhydrous $AlCl_3$ to give toluene. CH_3Cl in the presence of anhydrous $AlCl_3$ acts as an alkylation agent and introduces an alkyl group. This reaction is known as the Friedel-Crafts alkylation of benzene.

The reaction is as follows:



Thus, **option 2** is the correct answer.

Q62) Explanation:

Heterolysis is a type of chemical bond cleavage in which one of the bonded atoms gets both of the shared electrons. In homolysis, there is an equal distribution of the shared pair of electrons between two atoms to obtain two free radicals.

In propane, the electron density of methylene carbon atoms is greater than the electron density of terminal methyl groups. This is due to the electron donating (+I) effect of two methyl groups.

In the ethylium cation, the positive charge on methylene carbon atoms is stabilized through the electron-releasing (+I) effect of adjacent methyl groups.

The heterolysis of propane is given as: $\text{CH}_3\text{CH}_2\text{CH}_3 \rightarrow \text{CH}_3^- + \text{CH}_3\text{CH}_2^+$

Thus, **option 3** is the correct answer.

Q63) Explanation:

Hint: "A" is an aldehyde in the first step.

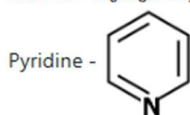
Q64) Explanation:

Hint: Compound must contain C and N

Nitrobenzene - $\text{C}_6\text{H}_5\text{NO}_2$

Hydroxyl Amine - H_2NOH

Aniline - $\text{C}_6\text{H}_5\text{NH}_2$



Q66) Explanation:

Hint: Ethyne has characteristic odour.

Explanation:

Alkynes are hydrocarbons which contain carbon-carbon triple bonds. Their general formula is $\text{C}_n\text{H}_{2n-2}$ for molecules with one triple bond (and no rings).

The first three members of alkyne family — ethyne, propyne and butyne are colourless gases, the next eight members are liquids and the higher ones are solids. Except for ethyne, which has characteristic smell, all members are odourless.

Alkynes are weakly polar in nature. They are lighter than water and immiscible with water but soluble in organic solvents like ethers, carbon tetrachloride and benzene.

So, **option 2** is the correct answer.

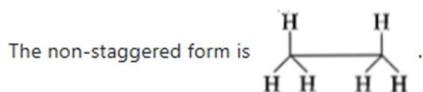
Q67) Hint: The non-staggered form(s) of ethane is called eclipsed form.

The alkane forms two conformations, that is, eclipsed and staggered. These conformations are formed because of free rotation around C-C single bond. These two conformations are interconvertible and obtained due to 60° rotation.

The conformation of ethane is as follows:



The non-staggered form(s) of ethane is called eclipsed form. Hence, option second is the correct answer.



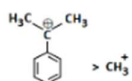
Q68) Explanation:

Hint: Benzyl carbocation is resonance stabilized.

Explanation:

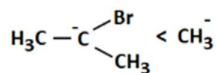
(1) Benzyl carbocation is more stable than methyl carbocation because benzyl carbocation is more stable as the positive charge on the carbon atom is resonance stabilized with the benzene ring.

Hyperconjugation is a factor in explaining why increasing the number of alkyl substituents on a carbocation or radical centre leads to an increase in stability.

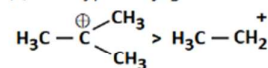


Thus, both hyperconjugation and resonance is responsible for the stability.

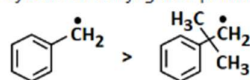
(2) The +I effect of methyl groups decreases the stability of carbanion. Although one bromine will stabilize the negative charge by -I effect. But two methyl groups are responsible for the less stability.



(3) The hyperconjugation and the inductive effect due to three methyl groups stabilizes the carbocation.



(4) Benzylic radicals are more stable than alkyl radicals because of resonance effects. An unpaired electron can be delocalized over a system of conjugated pi bonds. Thus, resonance is responsible for the stability.



Thus, the correct answer is **option 2**.

Q69) Explanation:

(i) A nucleophile is a chemical species that forms bonds by donating an electron pair. All molecules and ions with a free pair of electrons or at least one pi bond can act as nucleophiles.

Because nucleophiles donate electrons, they are Lewis bases. CN^- , H_2O , NH_3 and ROH are nucleophiles.

(ii) An electrophile is a chemical species that forms bonds with nucleophiles by accepting an electron pair. Because electrophiles accept electrons, they are Lewis acids.

H^+ , BF_3 , and $AlCl_3$ are electron-deficient species that act as an electrophile.

So, the correct set for nucleophiles is CN^- , H_2O , and ROH . Hence, **option 4** is the correct answer.

Q70) Explanation:

Hint: Electron withdrawing effect order is $-NO_2 > Cl > CH_3$

The electron-donating group increases the reactivity of the benzene ring towards EASR and the electron-withdrawing group decreases the electron density towards EASR.

CH_3 is electron-donating hence, c is the most reactive. The NO_2 group is the strong electron-withdrawing group hence, a is the least reactive. In the case of d and b, d is more reactive than b and less reactive than a.

Because Cl shows a strong -I effect and decreases the electron density of the benzene ring.

Hence, the reactivity order is $c > d > b > a$

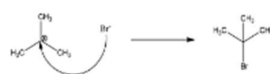
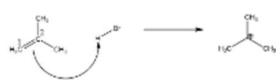
Q71) Explanation:

Hint: Tertiary carbocation is more stable than secondary carbocation.

Explanation:

(1) Protonation of an alkene: The formation of a tertiary carbocation takes place when an alkene reacts with HBr.

(2) Addition of Br^- to the carbocation leads to the formation of final product.

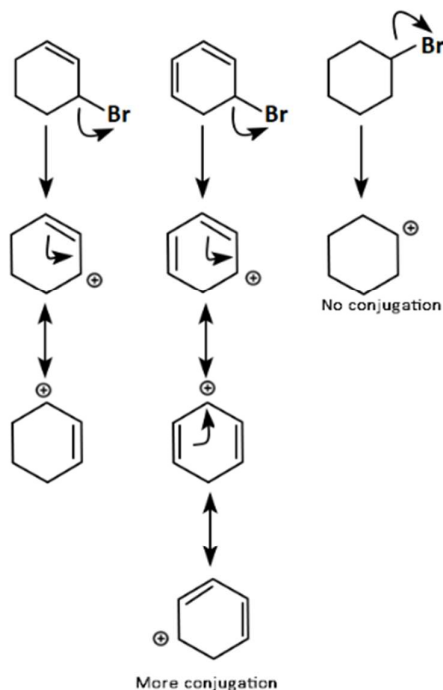


So, **option 1** is the correct answer.

Q72) Explanation:

The reaction of alkyl halide with strong base results in the formation of an alkene. In the transition state, the bromide ion will leave with a positive charge on the carbon atom.

So, the order of stability of carbocation will depend on the conjugation. Carbocations are stabilized by conjugation through pi bonds. This allows the positive charge to be delocalized through resonance.



Thus, the correct order of stability: II > I > III. So, **option 1** is the correct answer.

Q73) Explanation:

Types of Isomerism

1. **Tautomerism:** This usually involves the interconversion between two different forms of a compound, such as keto-enol tautomerism. For keto-enol tautomerism, a compound must have both a carbonyl group (keto form) and an alcohol group (enol form). For a compound with the formula $C_5H_{12}O$, there is no carbonyl group, so keto-enol tautomerism cannot occur. Examples of compounds with this formula are alcohols and ethers, which do not have a carbonyl group.
2. **Position Isomerism:** This type of isomerism involves different positions of the functional group on the carbon chain. For example, in alcohols or ethers with this formula, the position of the hydroxyl group or the alkyl groups can vary, leading to position isomers.
3. **Metamerism:** This occurs in compounds with the same functional group but different alkyl groups on either side. For ethers with this formula, different alkyl groups can give rise to metameric isomers.
4. **Functional Isomerism:** This involves different functional groups having the same molecular formula. For $C_5H_{12}O$, alcohols and ethers are functional isomers.

Conclusion

Given that the compound with $C_5H_{12}O$ does not include a carbonyl group, **keto-enol tautomerism** is not possible. Therefore, the correct conclusion is that this compound cannot exhibit tautomerism, especially keto-enol tautomerism, due to its lack of the necessary functional groups.

Hence, **option 1** is the correct answer.

Q74) Explanation:

Assertion (A):

"Propene reacts with HBr in the presence of H₂O₂ to give Markovnikov's addition product."

This statement is **true**. In the presence of H₂O₂, the reaction follows the **Markovnikov rule**, not the anti-Markovnikov rule. This is because H₂O₂ is a polar molecule and does not participate in the formation of free radicals. Radical formation typically requires organic peroxides, not H₂O₂. Hence, the reaction proceeds through the ionic mechanism, leading to the Markovnikov product.

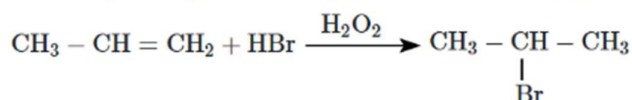
Reason (R):

"H₂O₂ forms free radicals which cause homolytic cleavage of H-Br."

This statement is **false**. H₂O₂ does not form free radicals in this reaction because it is polar and lacks the ability to initiate radical formation under these conditions. Free radicals are typically formed in the presence of organic peroxides, which facilitate the homolytic cleavage of H-Br. Therefore, the reason is incorrect in this context.

Conclusion:

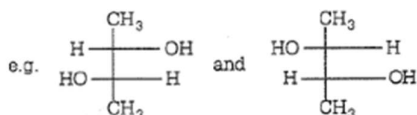
The correct option is **(3). Assertion (A) is True but Reason (R) is False.**



Q75) Explanation:

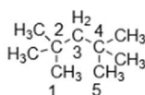
Hint: Enantiomers are mirror images of each other but they are not superimposable to each other.

Enantiomers are mirror images of each other but they are not superimposable to each other.



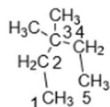
Q76) Hint: During numbering follow lowest sum rule

1. The first structure is as follows:



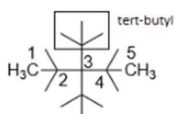
The parent chain contains five carbon atoms and 4 methyl groups. The parent chain name is pentane. The IUPAC name of the compound is 2,2,4,4-Tetramethylpentane.

2. The second structure is as follows:



The parent chain contains 5 carbon atoms. The two methyl groups are present in third position. Hence, dimethyl is used as a prefix. The correct IUPAC name of the compound is 3,3dimethylpentane.

(iii) The third structure is as follows:



The parent chain contains 5 carbon atoms. Hence, parent chain name is pentane. At third position, two tert-butyl group are present. At second and fifth position 4 methyl groups are present. Hence, the correct IUPAC name is 3,3-Di-tert-butyl -2,2,4,4-tetramethylpentane.

Q77) Explanation:

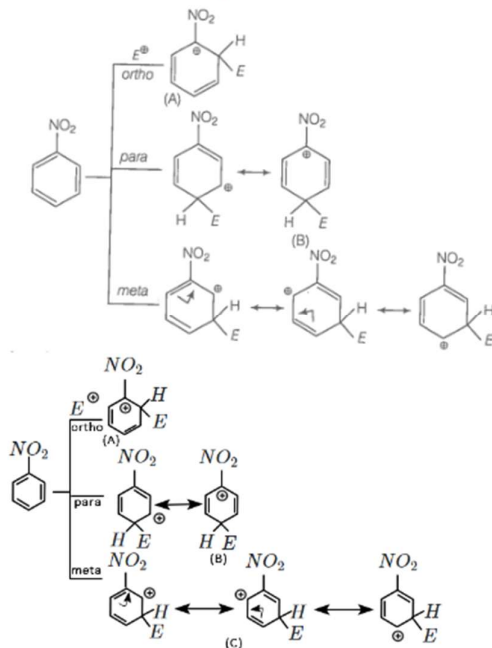
Hint: Nitro group is an electron-withdrawing effect.

Explanation:

Nitro group by virtue of -I effect withdraw electrons from the ring and increase the charged and destabilised carbocation.

In ortho, para-attack of electrophile on nitrobenzene, we are getting two structures (A) and (B) in which positive charge is appearing on the carbon atom directly attached in the nitro group.

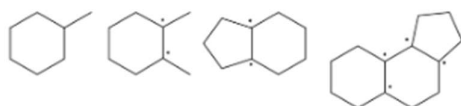
As nitro group is electron withdrawing by nature, it decreases the stability of such product and hence meta attack is more feasible when electron withdrawing substituents are attached.



Q78) Explanation:

Chiral carbon centers are carbon atoms that are attached to four different substituents and placed at the corners of a tetrahedron. Chiral carbon atoms are also referred to as 'stereogenic carbons' or 'asymmetrical carbon atoms'.

The chiral carbons in the given compounds are shown by the asterisk (*) mark:



Hence, **option 1** is the correct choice.

Q79) Hint: Electron donating group increases the reactivity of the benzene ring towards EASR.

Step 1:

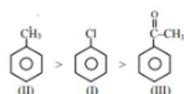
As the electron-donating group is attached to the benzene ring the electron density in the benzene ring increases and the reactivity of the benzene ring toward EASR increases. But in the case of the electron-withdrawing group reactivity of the benzene ring toward EASR decreases.

Step 2:

A CH_3 group is an electron-donating group hence it activates the benzene ring toward EASR. The COCH_3 group is a strong electron-withdrawing group it withdraws electrons by -R effect and deactivates the benzene ring towards EASR.

The chlorobenzene has intermediate reactivity because it shows a strong -I effect and weak +R effect. Hence, Cl is moderately deactivating.

Hence the reactivity order is as follows:



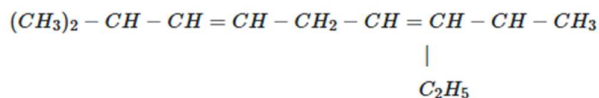
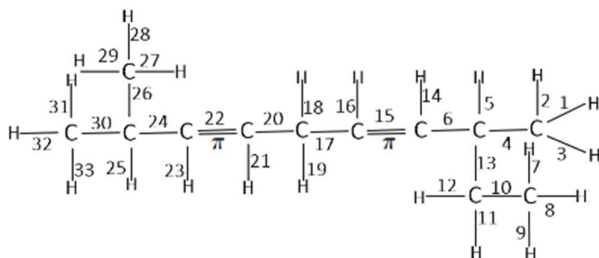
Q80) Explanation:

Hint: Double bond has one sigma and one pi bond.

Explanation:

Single bonds are comprised of one sigma bond whereas double bonds are comprised of one sigma and one pi bond. Triple bonds are comprised of one sigma bond and two pi bonds.

The sigma and pi bonds in the given structure are:

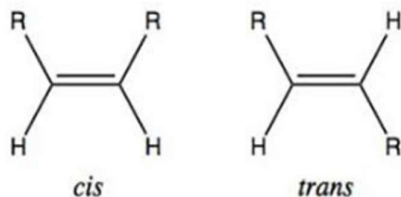


The given compound has 2π — bonds + 33σ — bonds. Thus, **option 1** is the correct answer.

Q82) Explanation:

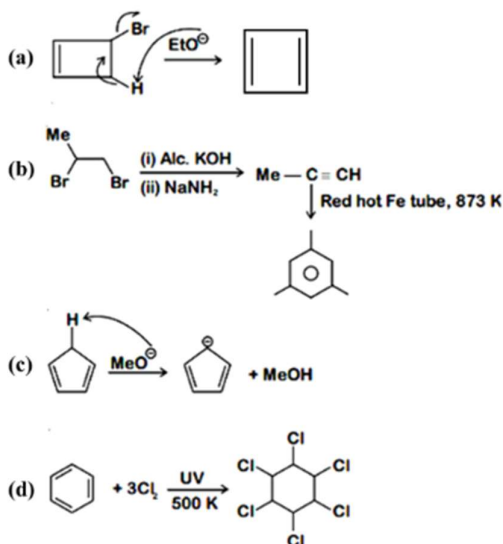
Geometrical isomers are two or more coordination compounds which contain the same number and types of atoms, and bonds (i.e., the connectivity between atoms is the same), but which have different spatial arrangements of the atoms.

e.g. cis- and trans- isomers are geometrical isomers.



Q83) Explanation:

Step 1: The given reaction proceeds as follows:



$C_6H_6Cl_6$ (Hexachlorocyclohexane):

- IUPAC Name: Hexachlorocyclohexane.
- Benzene hexachloride is **not an aromatic compound** as it does not have a delocalized pi-electron system.

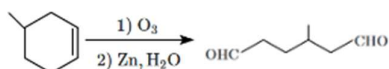
Q85) Explanation:

Hint: In ozonolysis, the carbon-carbon double bond in alkenes is cleaved.

Explanation:

Ozonolysis of alkenes cleaves the carbon-carbon double bonds and forms carbonyl compounds (aldehydes or ketones). The use of Zn and H_2O in the second step is a reductive workup, which prevents further oxidation of aldehydes to carboxylic acids.

The complete reaction is as follows:



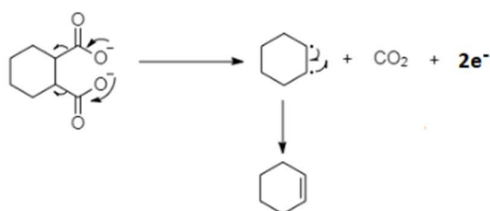
Hence, **option 1** is the correct answer.

Q86) Explanation:

double bond starts at **C2**

- **4-bromo**
 - **3-methyl**
 - **pent-2-ene**
- 4-Bromo-3-methylpent-2-ene

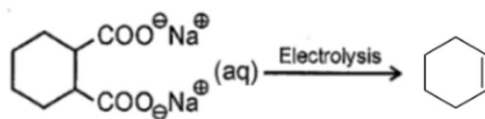
Q87) An aqueous solution of sodium or potassium salt of a carboxylic acid on electrolysis gives hydrocarbon. The reaction is as follows:



Step 2:

CO_2 will come out at the anode and H_2 at the cathode.

The reaction is as follows:



Q88) Explanation:

Hint: The (-I) effect refers to the electron-withdrawing inductive effect.

Explanation:

The -I effect (negative inductive effect) refers to the electron-withdrawing nature of a substituent or group attached to a molecule. It arises due to the electronegativity difference between the group and the atoms it is bonded to. The group pulls electron density through sigma bonds, creating a dipole and reducing electron density in the rest of the molecule.

This effect stabilizes positive charges or electron-deficient centers (e.g., carbocations) but destabilizes negative charges (e.g., carbanions). -I effect is caused by more electronegative atoms, e.g. NO_2 , Cl. Methyl and ethyl groups exert a +I effect.

Hence, **option 2** is the correct answer.

Q89) Explanation:

A pi bond formed by the overlap of p orbitals on adjacent atoms, perpendicular to any sigma bond(s) between the same atoms. So, the pi bond of the alkene have pure p-character.

Similarly, sigma bonds are more stable than pi bonds. Thus, pi-bonds are at a higher energy level than the electron pair in the σ -bond.

Hence, option 4 is the correct answer.

Q90) **Explanation:**

Hint: The bonding electron pair is split unevenly.

Explanation:

Heterolytic cleavage is a bond fission that occurs between two non-identical species. The most electronegative species acquires the electron pair from the covalent bond and becomes negatively charged. The other species becomes positively charged.

The electronegativity difference between oxygen and hydrogen is the greatest. Further, oxygen being highly electronegative can accommodate the negative charge more effectively developed after the cleavage.

So, **option 4** is the correct answer.